MILITARY SPECIFICATION
PRINTED WIRING BOARD

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of rigid printed wiring board entitled, Printed Wiring Board, consisting of a conductor pattern on the surface of two sides of an insulation base, with plated through holes, for use on the Lance warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-T-60	- Tape, Pressure Sensitive Adhesive, Waterproof, for Packaging and Sealing.
PPP-T-76	- Tape, Pressure Sensitive Adhesive, Paper, Water Resistant, for Carton Sealing.
L-T-90	- Tape, Pressure Sensitive Adhesive, (Cellophane and Cellulose Acetate).
PPP-B-566	- Box, Folding, Paperboard.
PPP-B-636	- Box, Fiberboard.
PPP-B-676	- Box, Setup.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-P-55110	- Printed Wiring Boards.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

NUOR

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MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-810 - Environmental Test Methods.

DRAWINGS

ORDNANCE CORPS

9317501 - Printed Wiring Board Detail.
9317500 - Printed Wiring Board.
EL-9317500 - Equipment List.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM).

ASTM-A-219 - Standard Methods of Test for Local Thickness of Electro-deposited Coatings.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The printed wiring board shall comply with all requirements specified on the drawing (dwg) 9317500, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated. The serial numbers shall be marked in accordance with MIL-STD-130.

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3.5 Plating adhesion. There shall be no evidence of any lifting or separation of platings from the surface of the conductor pattern or of the conductor from the base laminate. There shall be no whiskers or slivers of solder or plating on the surface of the printed wiring board. Testing shall be as specified in 4.4.1.

3.6 Circuit resistance. The resistance of each conductor pattern (see Table I) on the printed wiring board shall not be greater than 125 milliohms when tested as specified in 4.4.2.

Table I
Circuit resistance (see dwg 9317500)

Circuit Path	From Hole Number*	To Hole Numbers*
1	72	73, 272, 277
2	117	36,39,60,61,325,
3	29	28,377,380
4	118	273,276
5	172	382,387
6	151	150,314,319,333, 364,363
7	153	213
8	191	391,392
9	54	156
10	96	99
11	98	97,55
12	12	14
13	167	178,219,220
14	112	107
15	46	2
16	41	7
17	102	104
18	45	42
19	27	25
20	217	335
21	214	334
22	185	205
23	210	258
24	360	361
25	259	260
26	233	230
27	293	359
28	339	261
29	342	343
30	313	373
31	372	320
32	62	350
33	59	351
34	383	384

* - See *Note, Page 4.

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Table I (Cont'd)
Circuit resistance (see dwg 9317500)

Circuit Path	From Hole Number*	To Hole Number*
35	385	386
36	89	90
37	91	93
38	94	95
39	100	101
40	105	106
41	108	111
42	121	124
43	122	123
44	125	126
45	127	128
46	129	130
47	131	132
48	162	165
49	173	176
50	221	161, 166, 218
51	223	226
52	228	229
53	234	235
54	243	244
55	245	246

*Note: The numbers designated in the above table are keyed to the drilling numbers specified on dwg 9317501, sheets 2, 3 and 4 of 4.

3.7 Insulation resistance. The insulator resistance between each isolated conductor pattern and/or terminal area (see Table II) shall be not less than 500 megohms when 500 + 50 volts dc is applied for 10 seconds min, 60 seconds max, when tested as specified in 4.4.3. Current shall be limited to 5 milliamperes (ma).

Table II (See dwg 9317500)

Circuit Path	Positive Potential From Hole Number*	Negative Potential To Hole Number
1	72	71,74,120,273,279
2	117	119
3	29	43,379,114,172
4	-	-
5	172	173,243
6	151	152,149,320,120,313
7	153	154,212
8	191	190

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Table II (Cont'd)

Circuit Path	Positive Potential From Hole Number*	Negative Potential To Hole Number
9	54	59,155
10	96	97
11	-	-
12	12	13,34,38,15
13	167	165,177,176,161
14	112	111
15	46	45,3
16	41	42,6
17	102	101,93,130,105,127
18	45	48,41,49
19	27	29
20	-	-
21	214	215,213
22	185	153
23	210	217,235,259
24	360	359,362
25	-	-
26	233	234,229
27	-	-
28	339	357,293
29	342	354,325,352
30	313	375
31	372	370,278
32	59	120
33	114	50,59,172
34	161	153,172
35	350	72
36	382	383,386

*Note: The hole numbers designated in the above table are keyed to the drilling numbers specified on dwg 9317501 sheets 2, 3, 4 of 4. The polarity of the potential applied to the hole pairs shall be selected to minimize the time of testing.

3.8 Environmental testing. The printed wiring board shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The printed wiring board shall also comply with 3.6 and 3.7 before and after each environmental test. Testing shall be as specified in 4.4.4.1 and 4.4.4.2. Tests shall be conducted in the order specified.

3.8.1 Thermal shock. The printed wiring board shall withstand the shock of a temperature change from $160^{\circ} \pm 5^{\circ}F$ to $-65^{\circ} \pm 5^{\circ}F$ for five (5) cycles and shall show no visible evidence of deterioration, damage, or corrosion. Upon completion of this test the board shall meet the requirements of 3.6 and 3.7. Testing shall be as specified in 4.4.4.1.

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3.8.2 Moisture resistance. The printed wiring board shall be subjected to a 90 to 98 percent relative humidity and a temperature of 160° + 5°F for a minimum of 80 hours. There shall be no visible deterioration, including corrosion of the printed conductors or base material. Upon completion of this test the printed wiring board shall meet the requirements of 3.6 and 3.7. Testing shall be as specified in 4.4.4.2.

3.9 Plating thickness on circuit paths. The combined thickness of both clad and plated copper on the paths of the printed wiring board shall be a minimum of 0.004 inch. The thickness of the tin-lead plating on the paths of the printed wiring board shall be a minimum of 0.0005 inch. Testing shall be as specified in 4.4.5 for test coupon and for printed wiring board.

3.10 Tin-lead plating composition. The tin-lead plating shall contain a minimum of 50 percent and a maximum of 70 percent tin. Testing shall be as specified in 4.4.6 for test coupon and for printed wiring board.

3.11 Bond strength. A plated through hole shall withstand 20 pounds of pull after subjection to 5 cycles of unsoldering and shall not be loosened. Testing shall be as specified in 4.4.7.

3.12 Plating thickness of plated through holes. The thickness of the plating in the holes shall be 0.001 inches minimum for the copper plating and 0.0005 inches minimum for the solder plating and the measurement shall be as specified in 4.4.5. The following criteria shall also apply to the plating in the plated through holes:

a. There shall be no more than three (3) voids per hole. Total voids area shall not exceed 10 percent of the total wall area.

b. No plated hole shall have pits or voids at the junction of the hole wall and terminal area to a depth of 1 1/2 times the surface thickness. The surface thickness includes the copper cladding and the plating (copper and solder). Separation at the hole edge of the clad copper and plated copper is permissible unless the separation continues into the plated copper. See Figure 1 below:

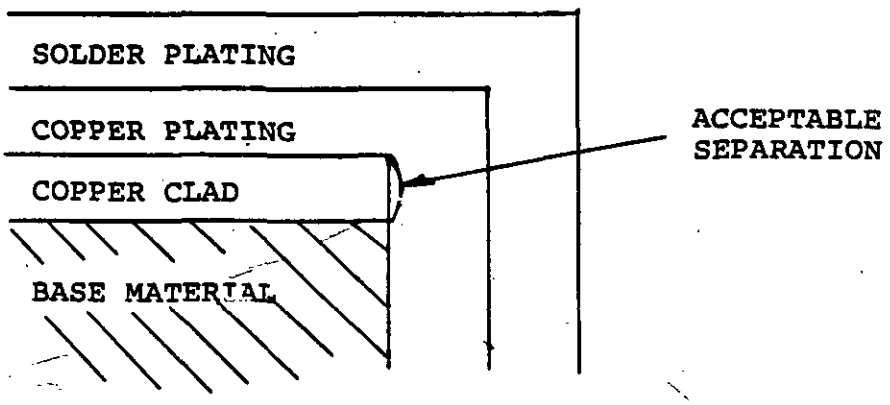


Figure 1

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3.13 Peel strength. The peel strength of the conductor path shall be a minimum of 11 pounds per inch of the effective conductor width at point of contact. Testing shall be as specified in 4.4.8

3.14 Solderability. All tin-lead plated surfaces of the printed wiring board must exhibit good solderability characteristics. Testing shall be as specified in 4.4.9.

3.15 Marking. Marking shall be in accordance with the master drawing and shall be made by the same process used in producing the conductor pattern. Marking shall be legible and shall in no case affect the performance of the printed wiring board.

3.16 Workmanship.

3.16.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.16.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.16.3 Burrs. Parts shall be free from burrs.

3.16.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, finger prints, mold release agent, flux residue or any embedded foreign material.

3.16.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 6 complete assemblies (printed wiring boards along with their companion test coupon for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table III. Tests shall be conducted in order specified in Table III.

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Table III

First article inspection and tests

Inspection or Test	PWB*	TC**	Dest	Non Dest
1. Tin-lead plating composition (see 3.10)	6	6		X
2. Classification of characteristics (see 4.3.2.1)	6	6		X
3. Plating adhesion (see 3.5)	6	6		X
4. Circuit resistance (see 3.6)	6			X
5. Insulation resistance (see 3.7)	6			X
6. Thermal shock (see 3.8.1)	6		X	
7. Moisture resistance (see 3.8.2)	6			X
8. Bond strength (see 3.11)	2		X	
9. Peel strength (see 3.13)	2		X	
10. Plating thickness of circuit paths (see 3.9)	6	6		X
11. Plating thickness of plated through holes (see 3.12)	2	6	X	

*PWB - Printed wiring board
**TC - Test coupon

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation. The formation of an inspection lot and related functions shall be in accordance with Section 5, MIL-STD-105. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table IV and Table V), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table IV
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table V

Minor Characteristics							
Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Printed wiring board (see dwgs 9317500 and 9317501).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Outside configuration of board	Gage
102.	Location of mounting holes	Gage
103.	Diameter of mounting holes	Gage
104.	Location of holes (see dwg 9317501)	Gage
105.	Diameter of holes (see dwg 9317501)	Gage
106.	Reduction in effective width of the conductor at any point shall not exceed 20 percent	Visual/Gage*
107.	Spacing between conductors greater than 0.010 inches	Visual/Gage*
108.	Conductor edge roughness less than 0.005 inches from peak to valley	Visual/Gage*
109.	No plated through hole may be any closer than 0.010 inches to the edge of its pad	Visual/Gage*

*The initial inspection for these characteristics shall be performed visually with the aid of a 4X magnifying device. Any apparent defect shall be further examined and measured, utilizing an approved measuring device, and compared with the above specified requirements.

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110. Plating must be present in all plated through holes.....Visual/Gage*
111. Configuration of conductor pattern.....Visual
112. Solder plating must be completely present on each pad.....Visual
113. No pits, nicks, scratches or cuts, exposing base material in conductor pattern or pads.....Visual
114. No blisters or wrinkles in conductor pattern or pads.....Visual
115. No separation of the plated through holes from the conductor pattern.....Visual
116. No separation of conductor pattern from the base material.....Visual
117. No delamination of base material.....Visual
118. No cracks, chips or bulges in base material.....Visual
119. Lack of tin-lead plating on any conducting surface.....Visual
120. Fibers extending into any plated through hole.....Visual
121. Any crack, chip or bulge in the board which weakens a conductor or is closer than 0.050 inch from a plated through hole.....Visual/Gage*
122. Cracks, pits or scratches in the tin-lead plating on the conductors which reveal the copper but do not extend through to the base material.....Visual
123. Objective evidence relating to material missing or improper (see 3.1).....Visual

*The initial inspection for these characteristics shall be performed visually with the aid of a 4X magnifying device. Any apparent defect shall be further examined and measured, utilizing an approved measuring device, and compared with the above specified requirements.

Minor:

201. Marking missing, unintelligible, misleading or incorrect location...Visual
202. Evidence of poor workmanship (see 3.16).....Visual

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4.3.2.1.2 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Packing and packaging not in accordance with applicable specification	Visual
102.	Marking not in accordance with MIL-STD-129	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.16)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test		Test Classification
a.	Plating adhesion	Major
b.	Circuit resistance	Major
c.	Insulation resistance	Major
d.	Tin-lead plating composition	Major
e.	Thermal shock	Major
f.	Moisture resistance	Major
g.	Bond strength	Major
h.	Peel strength	Major
i.	Plating thickness of plated through holes	Major
j.	Plating thickness of circuit paths	Major

Of these tests, e,g,h and i are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Plating adhesion, circuit resistance, insulation resistance and tin-lead plating composition. Each printed wiring board shall be tested in accordance with 4.4.1, 4.4.2, 4.4.3 and 4.4.6. If any item fails to comply with 3.5, 3.6, 3.7 and 3.10, the item shall be rejected and removed from the lot.

4.3.3.2 Thermal shock, moisture resistance, plating thickness of circuit paths, bond strength, peel strength and plating thickness of plated through holes. One sample printed wiring board shall be selected from every 25 printed wiring boards, or portion thereof, produced shall be tested in accordance with 4.4.4.1, 4.4.4.2, 4.4.5, 4.4.7 and 4.4.8. If

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any item fails to comply with 3.8.1, 3.8.2, 3.9, 3.11, 3.13 and 3.12, the 25 printed wiring boards from which the sample was selected shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-9317500). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Plating adhesion (see 3.5). A strip of pressure sensitive cellophane tape conforming to type 1, class A of L-T-90, 1/2 inch width and 2 inches long effective length, shall be pressed firmly across the surface of the conductor pattern. The tape shall then be removed rapidly by manual force applied approximately perpendicular to the pattern. The tape shall be applied to, and removed from, three different locations on each side of the printed wiring board tested. When contacts are part of the pattern at least one pull must be on contacts. Fresh tape shall be used for each test.

4.4.2 Circuit resistance (see 3.6). The circuit resistance test shall be performed utilizing an approved measuring device. The current shall be limited to 100 milliamps.

4.4.3 Insulation resistance (see 3.7). The insulation resistance test shall be performed utilizing and approved measuring device.

4.4.4 Environmental testing (see 3.8).

4.4.4.1 Thermal shock (see 3.8.1). Printed wiring boards shall be tested in accordance with Method 107 of MIL-STD-202. The following details and exceptions shall apply:

a. Test conditions shall be as follows:

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Temperature (°F)	Time (minutes)
-65 ± 5	30 min
+70 ± 10	5 max
160 ± 5	30 min
70 ± 10	5 max

b. Examination after test. The board shall be visually examined for evidence of deterioration, damage, and corrosion. Upon completion of this test, the board shall be tested in accordance with 4.4.2 and 4.4.3.

4.4.4.2 Moisture resistance (see 3.8.2). Printed wiring boards shall be tested in accordance with Method 106 of MIL-STD-202. The following details shall apply:

a. Omit steps 7A and 7B.

b. The temperature of the chamber shall be 160 + 5°F during the humidity cycles.

c. Final measurements - Following step 6 of the final cycle, the printed wiring board shall be removed from the humidity chamber. Remove the surface moisture and commence taking measurements of insulation resistance, (see 4.4.3) within one minute after removal of the printed wiring board from the humidity chamber.

4.4.5 Plating thickness on circuit paths (see 3.9) and in plated through holes (see 3.12).

4.4.5.1 Test coupon. Two sections, each containing six plated through holes shall be removed from coupon. The mounting of these specimens shall be in an appropriate plastic material utilizing a casting technique which does not require heat or pressure. The thickness shall be measured in accordance with ASTM-A-219. Magnification of 44x or greater shall be used for all thickness readings. Thickness readings on circuit paths shall be made at a point approximately midway between the plated through holes. Thickness readings in the plated through holes shall be made at points approximately in the middle of the hole.

4.4.5.2 Printed wiring board. Three sections, each containing three plated through holes and at least one connecting land shall be removed form the printed wiring board. Each of these sections shall contain two different size plated through holes. These specimens shall be mounted and measured in the same manner described in 4.4.5.1.

4.4.6 Tin-lead plating composition. The composition shall be measured using a Betascope (see 6.3) or equivalent and in accordance with the manufacturers operating manual.

4.4.6.1 Test coupon. The two 5/16" diameter pads on coupon shall be utilized in determining this value. Both readings must fall within the requirement.

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4.4.6.2 Printed wiring board. Six different land areas, three on each side of the printed wiring board shall be utilized in determining this value. All readings must fall within the requirement.

4.4.7 Bond strength (3.11). A minimum of five (5) randomly selected plated through holes of each hole size shall be used for this test. The bond strength test shall be performed within three hours after removal of specimen from humidity chamber and temperature of specimen shall not exceed room temperature. Take a sharp model knife and cut conductor at least 1/4 inch away from terminal areas selected for bond strength test. Start conductor peeling toward terminal area. Place blade of knife on edge of terminal area where conductor enters. While holding blade down, peel conductor from terminal area and use knife blade as fulcrum to break conductor from terminal area without disturbing bond of the terminal area. Select suitable wire of sufficient length to connect to gripping mechanism of bond strength tester. Insert wire in holes in selected terminal areas and solder to terminal areas by machine or hand, as applicable. The wires shall not be clinched. Subject wires to five cycles of unsoldering and soldering by hand after the initial machine or hand soldering. During the five cycles the wires shall be completely removed during each unsoldering operation and replaced during each soldering operation. A 60-watt conventional soldering iron operating at a reduced voltage sufficient to produce a tip temperature of 450°F to 500°F shall be used for the unsoldering and soldering operation. The iron shall be applied to the leads, not the foil, and shall be applied only as long as is necessary to perform the unsoldering of soldering operation. Following the fifth cycle, the printed wiring board shall then be clamped in the jaws of the bond tester. A pull at the rate of 2 inches per minute shall be applied to the wire on the pattern side of the board. The load shall be applied perpendicular to the major surface of the terminal area until the required poundage is reached or a failure occurs. Breaking of a wire, or wire pullout shall not be considered as a failure, but the wire shall be resoldered and pulled again. It shall be considered a failure when one of the following conditions occurs:

a. A plated hole is loosened with soldered wire attached, whether or not the terminal is also loosened.
b. A terminal area around a bare hole is loosened.

4.4.8 Peel strength (see 3.13). A sample of three (3) randomly selected conductor paths on the board shall be used for this test. The conductor paths shall be a straight length of a minimum of three (3) inches. The conductor path shall be peeled back to approximately one (1) inch so that the line of pee. is perpendicular to the edge of the board. Each board shall be clamped or held on a horizontal surface with the peeled

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conductor path projecting upward for one (1) inch. The end of the conductor path shall be gripped between two knurled jaws of a clamp with a chain attachment to a dial, indicating-force indicator or a tensile tester which has been adjusted to compensate for the weight of the chain and clamp. The jaws shall cover the full width of the conductor path and shall be exerted in a vertical plane and shall not deviate from the perpendicular by more than 5°. The conductor path shall be pulled from the base material at a speed of 2 ± 1/10 inches per minute; the minimum load shall then be observed, converted to pounds-per-inch width and shall be considered as the peel strength. If the full width of the conductor path does not peel, the result shall be discarded and another conductor path shall be tested. The peel strengths are the average for all conductors tested on one side of the board. For the test coupon start from the wide end of each of the four (4) specimens (2 on bottom and 2 on top).

4.4.9 Solderability (see 3.14). The solderability pattern on the test coupon shall be used for this test and shall be done in accordance with MIL-P-55110, 4.9.11.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Printed wiring boards shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Printed wiring boards shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packaging. Unless otherwise specified, printed wiring boards shall be individually packaged in accordance with MIL-P-116, Method IA8, insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein. Bagged boards, with any dimension larger than 6 inches, shall be placed in a supplementary container conforming to PPP-B-636, class weather resistant, style optional, special requirement use. The closure, waterproofing, and banding requirements shall conform to the procedure specified in 5.2.1. The unit packaging may serve as a shipping container.

5.1.1.5 Intermediate packaging. Printed wiring boards, packaged as described in 5.1.1.4, when the container is not already a carton, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size and shape, shall be of minimum tare and cube, and shall contain multiples of five unit packages, not to exceed 100 packages and ten pounds. No intermediate packaging is required when the total quantity shipped to a single destination is less than 100 units.

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5.1.2 Level C. Each printed wiring board shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, or C, as specified (see 6.1).

5.2.1 Level A. The packaged printed wiring boards shall be packed in fiber-board containers conforming to PPP-B-636, class weather resistant, style optional, special requirement. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closures and waterproofing shall be accomplished by sealing all seams, corners, and manufacturer's joints with waterproof tape, 2 inches minimum width conforming to PPP-T-60, class 1, or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636, using non-metallic or tape banding only.

5.2.2 Level C. The packaged printed wiring boards shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package, intermediate and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

Custodian:
Army - PA

Preparing activity:
Army - PA

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